

Digital Filters

Physical Measuring Techniques FK7063

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Introduction

In this computational lab, we construct several digital filters using Matlab and compare them to each other. We also analyze data from an EKG signal and determine ways to remove noise from it.

1 Tasks

1.1 Constructing a Second Order Stop-Band Filter

We define the frequency response of a stop-band filter whose gain is 0 at $f = f_0 = 50$ Hz with a bandwidth of $b = 4$ Hz at -3 dB (corresponding to an attenuation of $1/\sqrt{(2)}$). The sampling rate is $F_s = 1000$ Hz.

In the space of the z-transform, the frequency f_0 corresponds to $\exp(i\Omega_0)$ with

$$\Omega_0 = 2\pi \frac{f_0}{F_s} = \frac{\pi}{10}$$

To get the desired attenuation, we place a zero at this point, i.e. we add a factor $(1 - \exp(i\Omega_0))$ to the denominator. Since real signals always contain negative frequencies in the same amount as corresponding positive frequencies, we also need to attenuate -50 Hz (corresponding to $-\Omega_0$) in the same way, so we add a factor $(1 - \exp(-i\Omega_0))$.

Now we need to add the property that the filter's maximum gain is 1 and that it has the desired bandwidth. This is achieved by adding a pole (i.e. a factor in the denominator) that is not on the unit circle but still close to the zero that we added. It can be parametrized as $r \exp(i\Omega_0)$ (with r real). For values of z that are far away, the distance to the pole and the zero will be roughly the same, so modulus of the numerator and denominator are equal, leading to a gain of 1. When z approaches the pole and the zero, there comes a point when it is separated from the zero by a distance $d \approx (1 - r)$ and from the pole by a distance $\approx \sqrt{d + (1 - r)} \approx \sqrt{2}(1 - r)$. the ratio of the numerator and denominator is then $1/\sqrt{(2)}$, so the arc around the unit circle where the gain is lower

than -3 dB has a length of $2d \approx 2(1 - r)$. The bandwidth of the filter thus depends on the distance r of the pole to the origin. We require the arc $2d$ to correspond to the bandwidth $b = 4\text{ Hz}$:

$$\begin{aligned} 2(1 - r) &= 2d \stackrel{!}{=} 2\pi \frac{b}{F_s} \\ \implies r &= 1 - \frac{b}{F_s} \pi = 1 - \frac{\pi}{250} = 0.98743 \end{aligned}$$

Again, we add two poles for positive and negative frequencies. With both zeros and poles, the filter transfer function becomes

$$H(z) = \frac{(1 - \exp(i\Omega_0))(1 - \exp(-i\Omega_0))}{(1 - r \exp(i\Omega_0))(1 - r \exp(-i\Omega_0))}$$

Multiplying out the brackets and then dividing through by z^2 , we get $H(z)$ as a ratio of polynomials in z^{-1} :

$$\begin{aligned} H(z) &= \frac{B(z)}{A(z)} = \frac{1 - 2\cos(\Omega_0) \cdot z^{-1} + z^{-2}}{1 - 2r\cos(\Omega_0) \cdot z^{-1} + r^2 \cdot z^{-2}} \\ &= \frac{1 - 1.9021z^{-1} + z^{-2}}{1 - 1.8782z^{-1} + 0.9750z^{-2}} \end{aligned}$$

This is the form used by the Matlab function `freqz(b, a, N)` to plot the transfer function. b and a specify the coefficients in $B(z)$ and $A(z)$ so $\mathbf{b} = [1 \ -1.9021 \ 1]$ and $\mathbf{a} = [1 \ -1.8782 \ 0.9750]$. This gives the plot shown in Fig. 1

The code is shown in Fig. 2

1.2 Second Order Butterworth-Type Stop-Band Filter

Next we construct a stop-band filter with the same characteristics as before, but created as a Butterworth filter using the Matlab function `butter(n, Wn, 'stop')`. $\mathbf{n}$ is the order where $\mathbf{n} = 1$ actually corresponds to a second order filter, and $\mathbf{Wn}$ is an array containing the break points normalized to the Nyquist frequency. `'stop'` specifies that it is a stop-band filter.

The transfer function plot can be seen in Fig. 3 and the code in Fig. 4.

The transfer functions of the previous filter and the butterworth-type filter have very similar characteristics since they were both defined to have the same stop band with the same bandwidth. Both filters have a jump in phase around the center of the stop-band. The manually defined filter drops more sharply (in theory reaching $-\infty\text{ dB}$) in the center of the stop-band because it has a zero there, whereas the butterworth-type filter has a finite minimum.

2 Conclusion

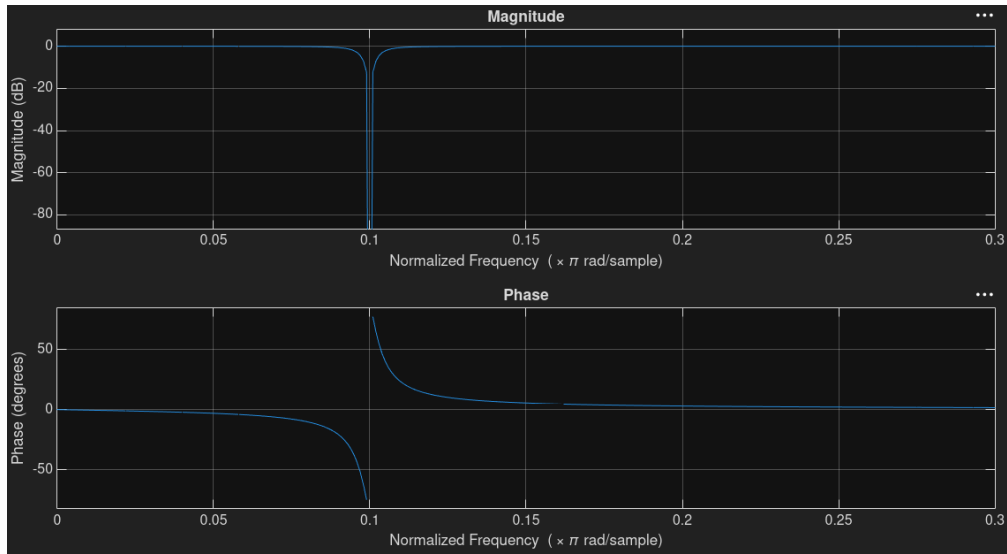


Figure 1: Plot of the transfer function of the second-order stop-band filter, created using the Matlab function `freqz`. The frequency range was chosen to zoom in on the relevant features. We chose 1000 samples for better visibility.

```
% input parameters
Fs = 1000;
f0 = 50;
b = 4;

% calculate omega0 and r
omega0 = f0/Fs * 2*pi;
r = 1-b/Fs*pi;

% calculate polynomial coefficients
b = [1 -2*cos(omega0) 1];
a = [1 -2*r*cos(omega0) r^2];

% plot using freqz
freqz(b, a, 256);
xlim([0 0.3]);
subplot(2, 1, 2); xlim([0 0.3]);
```

Figure 2: Matlab code to define and plot the second order stop-band filter.

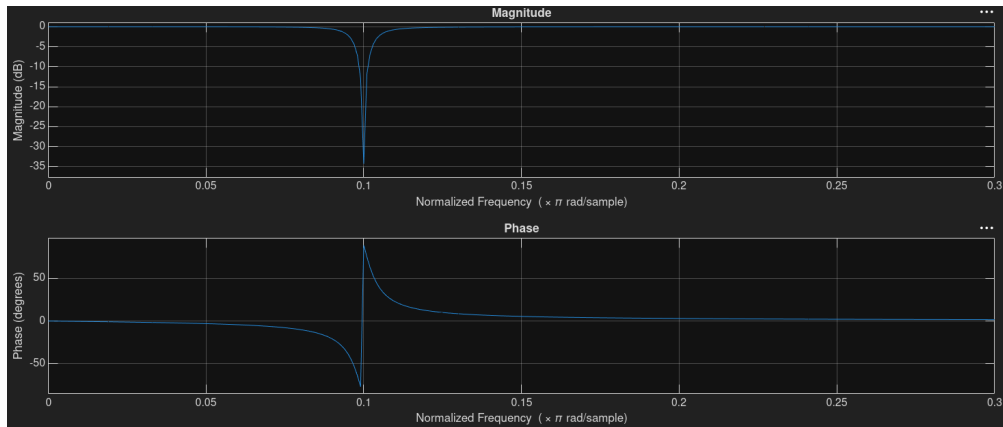


Figure 3: Plot of the transfer function of the second-order butterworth-type stop-band filter.

```
% input parameters
Fs = 1000;
f0 = 50;
b = 4;

% generate filter
n = 1;
Wn = [f0-b/2 f0+b/2] / (Fs/2);
[b, a] = butter(n, Wn, 'stop');

% plot transfer function
freqz(b, a, 1000)
xlim([0 0.3]);
subplot(2, 1, 2); xlim([0 0.3]);
```

Figure 4: Matlab code to generate and plot a second order butterworth-type stop-band filter.